29 The ratio of ^{14}C : ^{12}C in living things is fixed when they are alive but decreases after death because of the radioactive decay of ^{14}C. ^{14}C has a half-life of 5730 years. A 10.0 g sample of wood cut from a live tree is measured to have an activity of 120 min^{-1}.

If a 1.0 g sample of an old wooden relic is found to have an activity of 8.05 min^{-1}, how long ago was this wooden relic cut from a live tree?

A 1470 years **B** 3300 years **C** 9950 years **D** 22300 years

Ans: **B**

Since $A \propto N$,

$$A_{1.0g\ \text{live wood}} = \frac{120}{10} = 12.0\ \text{min}^{-1}$$

$$A = A_0 e^{-\lambda t}$$

$$8.05 = 12.0 e^{-\frac{\ln 2}{5730} t}$$

$$t = 3300\ \text{years}$$

30 Uranium-235 undergoes fission in a reaction shown below, releasing 195 MeV of energy. The